Yuxi Chen

yxch5277@mail.ustc.edu.cn

School of Mathematical Sciences, University of Science and Technology of China (USTC)

Education

Sep. 2026 – Present **Ph.D. Student in Mathematics**, School of Mathematical Sciences, University of Science and Technology of China (USTC). Expected completion: 2031.
Supervisor: Prof. Juyong Zhang.

Sep. 2022 – Jun. 2026 **B.Sc. in Mathematics**, major in Information and Computing Science, University of Science and Technology of China (USTC).

Research Experience

Sep. 2026 – Present **High-Fidelity 3D Facial Reconstruction and Expression Deformation Modeling**
University of Science and Technology of China

- Working on finding a high-fidelity 3D facial representation.
- Investigating facial deformation modeling for detailed 3D face representation.
- Conducting experiments on mesh optimization and reconstruction quality evaluation.

Selected Projects

Oct. 2025 – May. 2026 **Audio-driven emotion control for single image for video generation**
Undergraduate Thesis | *University of Science and Technology of China*
Supervisor: Prof. Juyong Zhang

- Contributed to an existing two-stage audio-driven avatar generation framework, where driving audio is mapped to FLAME expression coefficients for reference-image-conditioned video synthesis.
- Integrated SyncNet-based audio-visual synchronization feedback into the fine-tuning process of the upstream FLAME-to-pixel prediction network, exploring an auxiliary training signal for lip-synchronization refinement.

May. 2025 **Incremental Sparse Structure-from-Motion (SfM) Reconstruction**
Mathematical Modeling Course Project | *University of Science and Technology of China*

- Implemented camera calibration, SIFT feature extraction, FLANN matching, and RANSAC-based relative pose estimation for sparse 3D reconstruction.
- Developed triangulation, incremental camera registration with PnP-RANSAC, and Ceres-based bundle adjustment; visualized reconstructed point clouds using Open3D.

Research Interests

- 3D Facial Geometry Reconstruction and Deformation Modeling
- 3D Human Representation and Controllable Avatar Generation
- Generative Models for Human and Video Generation
- Multimodal AI and Interactive Avatars

Selected Coursework

GRADUATE COURSEWORK

- Digital Image Processing 91/100
- Geometric Deep Learning 85/100
- Numerical Methods for PDEs 92/100

UNDERGRADUATE COURSEWORK

- Computer Graphics 94/100
- Frontiers of Computer Graphics 94/100
- Mathematical Modeling 4.0/4.3
- Probability Theory 92/100
- Mathematical Statistics 95/100

Awards & Honors

2025	Silver Award, USTC Outstanding Student Scholarship
2022	USTC Outstanding Freshman Scholarship